

ELITE MATHEMATICS

Pearson Edexcel International GCSE

# Linear 4MA1

## Adaptive Classified

OL-T51 · Vectors

### With Answers

Questions with full worked solutions

■ 44 pages

# FAMILY MAP

## Vectors

|            |                                                  |                 |
|------------|--------------------------------------------------|-----------------|
| <b>F01</b> | <b>Use vectors with ratios and intersections</b> | 14 Q   67 marks |
|------------|--------------------------------------------------|-----------------|

|            |                               |               |
|------------|-------------------------------|---------------|
| <b>F02</b> | <b>Calculate with vectors</b> | 2 Q   5 marks |
|------------|-------------------------------|---------------|

|            |                                   |               |
|------------|-----------------------------------|---------------|
| <b>F03</b> | <b>Read and represent vectors</b> | 1 Q   1 marks |
|------------|-----------------------------------|---------------|

23

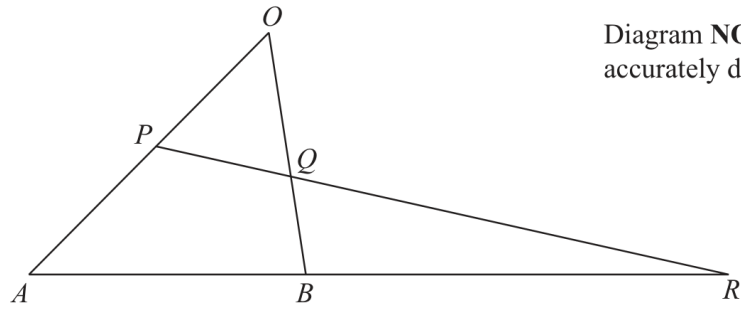


Diagram **NOT**  
accurately drawn

$OAB$  is a triangle.

$P$  is the midpoint of  $OA$

$Q$  is a point on  $OB$

$ABR$  and  $PQR$  are straight lines.

$$\vec{OA} = 12\mathbf{a} \quad \vec{OB} = 8\mathbf{b}$$

(a) Express  $\vec{AB}$  in terms of  $\mathbf{a}$  and  $\mathbf{b}$

(1)

**Continued**  
4MA1-1H Q23

## ORIGINAL QUESTION

23

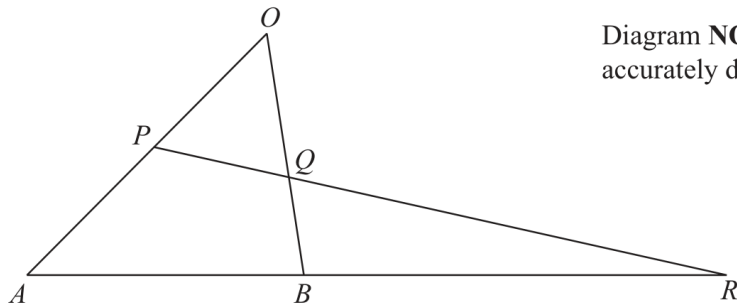


Diagram **NOT**  
accurately drawn

$OAB$  is a triangle.

$P$  is the midpoint of  $OA$

$Q$  is a point on  $OB$

$ABR$  and  $PQR$  are straight lines.

$$\vec{OA} = 12\mathbf{a} \quad \vec{OB} = 8\mathbf{b}$$

$$AB:BR = 1:2 \quad \vec{OQ} = n\mathbf{b}$$

(b) Use a vector method to find the value of  $n$

$$n = \dots\dots\dots (4)$$

**Worked Solution 14**

Vector method on two straight lines

**UNIT 2**  
**5 marks****Find AB**

$$\vec{AB} = \vec{OB} - \vec{OA} = 8\mathbf{b} - 12\mathbf{a}.$$

**Find PR**

$$\vec{OR} = \vec{OA} + 3\vec{AB} = -24\mathbf{a} + 24\mathbf{b}, \quad \vec{OP} = 6\mathbf{a}, \quad \vec{PR} = -30\mathbf{a} + 24\mathbf{b}.$$

**Use collinearity**

$$\vec{OQ} = \vec{OP} + \lambda\vec{PR} = 6\mathbf{a} + \lambda(-30\mathbf{a} + 24\mathbf{b}). \quad 6 - 30\lambda = 0 \Rightarrow \lambda = \frac{1}{5}, \quad n = 24\lambda = \frac{24}{5}.$$

**Final answer**

$$n = \frac{24}{5}$$

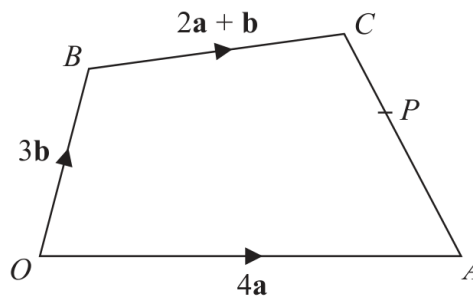
**Question 26**

4MA1-2H Q24

ORIGINAL QUESTION

6 marks

24

Diagram **NOT**  
accurately drawnThe diagram shows a quadrilateral  $OACB$  in which

$$\vec{OA} = 4\mathbf{a} \quad \vec{OB} = 3\mathbf{b} \quad \vec{BC} = 2\mathbf{a} + \mathbf{b}$$

- (a) Find  $\vec{AC}$  in terms of  $\mathbf{a}$  and  $\mathbf{b}$   
Give your answer in its simplest form.

$$\vec{AC} = \dots\dots\dots (2)$$

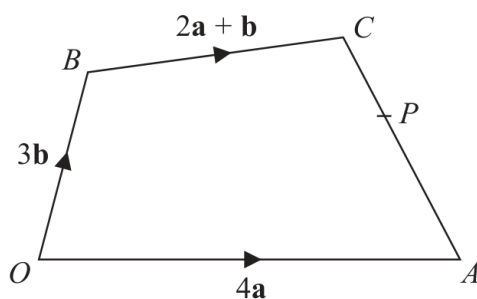
**Question 26 (continued)**

4MA1-2H Q24

ORIGINAL QUESTION

6 marks

24

Diagram **NOT**  
accurately drawn

The diagram shows a quadrilateral  $OACB$  in which

$$\vec{OA} = 4\mathbf{a} \quad \vec{OB} = 3\mathbf{b} \quad \vec{BC} = 2\mathbf{a} + \mathbf{b}$$

The point  $P$  lies on  $AC$  such that  $AP:PC = 3:2$

The point  $Q$  is such that  $OPQ$  and  $BCQ$  are straight lines.

- (b) Using a vector method, find  $\vec{OQ}$  in terms of  $\mathbf{a}$  and  $\mathbf{b}$   
Give your answer in its simplest form.  
Show your working clearly.

$$\vec{OQ} = \dots\dots\dots (4)$$

**Worked Solution 26**

Vectors and collinearity

MODULAR UNIT 2

6 marks

**Part (a)**

$$\vec{AC} = 4b - 2a.$$

**Point P**

$$\vec{OP} = 4a + \frac{3}{5}(4b - 2a) = \frac{14}{5}a + \frac{12}{5}b.$$

**Point Q**

$$\vec{OQ} = \lambda \vec{OP} = 3b + \mu(2a + b) \Rightarrow \lambda = 3.$$

**Final answer**

$$\vec{OQ} = \frac{42}{5}a + \frac{36}{5}b$$

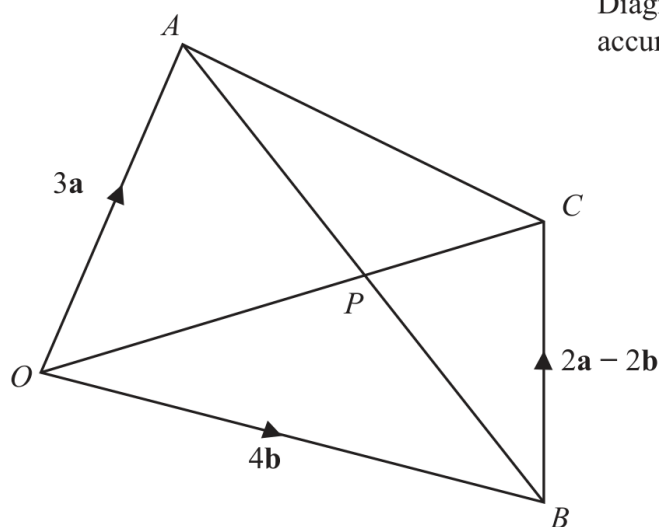
**Question 35**

4MA1-2H Q22 (a)(i), (a)(ii), (b)

ORIGINAL QUESTION UNIT 2

5 marks

22

Diagram **NOT**  
accurately drawn $OACB$  is a quadrilateral.

$$\vec{OA} = 3\mathbf{a} \quad \vec{OB} = 4\mathbf{b} \quad \vec{BC} = 2\mathbf{a} - 2\mathbf{b}$$

- (a) (i) Find the vector  $\vec{OC}$  in terms of  $\mathbf{a}$  and  $\mathbf{b}$   
Simplify your answer.

$$\vec{OC} = \dots\dots\dots$$

(1)

- (ii) Find the vector  $\vec{AB}$  in terms of  $\mathbf{a}$  and  $\mathbf{b}$

$$\vec{AB} = \dots\dots\dots$$

(1)

The point  $P$  lies on  $AB$  and on  $OC$ 

- (b) Using a vector method, find the ratio  $AP : PB$   
Show your working clearly.

$$\dots\dots\dots$$

(3)

**Worked solution 35****MODULAR UNIT 2**Find OC from two vectors / Find AB from position vectors / Ratio where two vector descriptions meet **5 marks****(a)(i) – Find OC from two vectors**

Use the path O-B-C

$$\overrightarrow{OC} = \overrightarrow{OB} + \overrightarrow{BC} = 4b + (2a - 2b) = 2a + 2b.$$

**Final answer**

$$OC = 2a + 2b$$

**(a)(ii) – Find AB from position vectors**

Subtract position vectors

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = 4b - 3a.$$

**Final answer**

$$AB = 4b - 3a$$

**(b) – Ratio where two vector descriptions meet**

Describe P on both lines

$$\overrightarrow{OP} = \lambda(2a + 2b) = 3a + \mu(4b - 3a).$$

Compare coefficients

$$2\lambda = 4\mu, \quad 2\lambda = 3 - 3\mu \Rightarrow \mu = \frac{3}{7}.$$

Convert the parameter to a ratio

$$AP : PB = \mu : (1 - \mu) = \frac{3}{7} : \frac{4}{7} = 3 : 4.$$

**Final answer**

$$AP:PB = 3:4$$

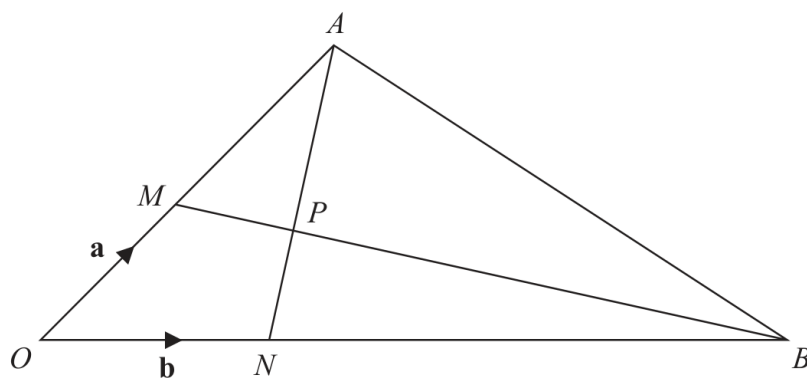
**Question 15**

4MA1-1H Q19

ORIGINAL QUESTION

5 marks

19

Diagram **NOT**  
accurately drawn

$OMA$ ,  $ONB$ ,  $MPB$  and  $NPA$  are straight lines.

$M$  is the midpoint of  $OA$

$ON:NB = 1:5$

$$\vec{OM} = \mathbf{a} \quad \vec{ON} = \mathbf{b}$$

(a) Find in terms of  $\mathbf{a}$  and  $\mathbf{b}$  the vector  $\vec{AN}$

.....  
(1)

(b) Use a vector method to find the ratio  $AP:PN$

$AP:PN =$  .....  
(4)

**Worked Solution 15**

Intersection ratio by vectors

MODULAR UNIT 2

5 marks

**Write the key vectors**

$$\overrightarrow{OA} = 2\mathbf{a}, \quad \overrightarrow{OB} = 6\mathbf{b}, \quad \overrightarrow{AN} = \mathbf{b} - 2\mathbf{a}.$$

**Describe P along AN**

$$\overrightarrow{OP} = 2\mathbf{a} + \lambda(\mathbf{b} - 2\mathbf{a}) = (2 - 2\lambda)\mathbf{a} + \lambda\mathbf{b}.$$

**Describe P along MB**

$$\overrightarrow{OP} = \mathbf{a} + \mu(6\mathbf{b} - \mathbf{a}) = (1 - \mu)\mathbf{a} + 6\mu\mathbf{b}.$$

**Equate coefficients**

$$\lambda = 6\mu, \quad 2 - 2\lambda = 1 - \mu \Rightarrow \lambda = \frac{6}{11}.$$

**Final answer**

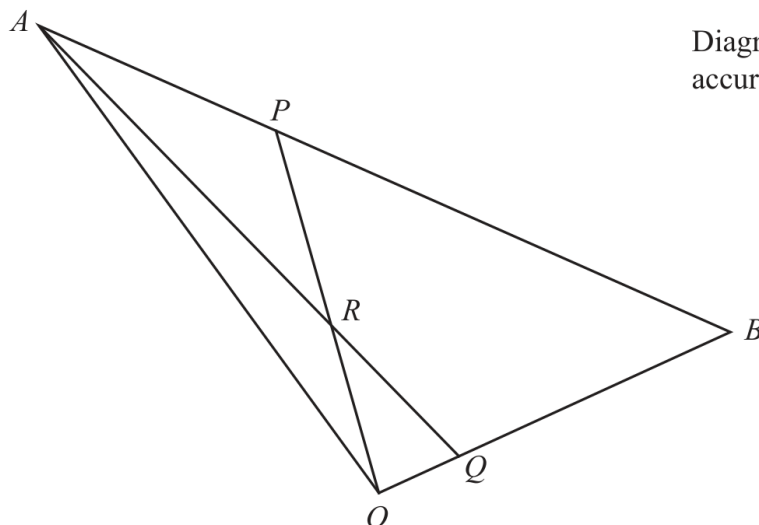
$$\overrightarrow{AN} = \mathbf{b} - 2\mathbf{a}; \quad AP : PN = 6 : 5$$

**Original question 16**

4MA1-1H Q24 M01, M02, M03

MODULAR UNIT 2

6 marks

24  $OAB$  is a triangle.Diagram **NOT**  
accurately drawn

$$\vec{OA} = 10\mathbf{a} \quad \vec{OB} = 10\mathbf{b}$$

 $ARQ$  and  $ORP$  are straight lines.

$$\vec{AP} = \frac{1}{4} \vec{AB} \quad \text{and} \quad \vec{OQ} = \frac{1}{5} \vec{OB}$$

Write the following vectors in terms of  $\mathbf{a}$  and  $\mathbf{b}$   
Simplify your answers.

(i)  $\vec{AQ}$

.....  
(1)

(ii)  $\vec{OP}$

.....  
(1)

**Original question 16 (continued)**

4MA1-1H Q24 M01, M02, M03

MODULAR UNIT 2

6 marks

(iii)  $\vec{OR}$ 

---

  
(4)

**Worked solution 16**

Find AQ / Find OP / Find OR from two straight-line routes

MODULAR UNIT 2

6 marks

**M01 – Find AQ****Use a route through O**

$$\overrightarrow{AQ} = \overrightarrow{AO} + \overrightarrow{OQ} = -10\mathbf{a} + 2\mathbf{b}.$$

**Final answer**

$$-10\mathbf{a} + 2\mathbf{b}$$

**M02 – Find OP****Use AP=AB/4**

$$\overrightarrow{AP} = \frac{1}{4}(10\mathbf{b} - 10\mathbf{a}) = \frac{5}{2}\mathbf{b} - \frac{5}{2}\mathbf{a}.$$

**Add OA**

$$\overrightarrow{OP} = 10\mathbf{a} + \overrightarrow{AP} = \frac{15}{2}\mathbf{a} + \frac{5}{2}\mathbf{b}.$$

**Final answer**

$$\frac{15}{2}\mathbf{a} + \frac{5}{2}\mathbf{b}$$

**M03 – Find OR from two straight-line routes****Write OR through P**

$$\overrightarrow{OR} = \overrightarrow{OP} + \lambda\overrightarrow{PR}.$$

**Write OR through Q**

Use the line ARQ and the known AQ route to form a second expression.

**Equate coefficients**The two routes give  $\lambda=1/2$  and hence vector  $\overrightarrow{OR}=(15)/(4)\mathbf{a}+5/4\mathbf{b}$ .**Simplify**

$$\overrightarrow{OR} = \frac{15}{4}\mathbf{a} + \frac{5}{4}\mathbf{b}.$$

**Final answer**

$$\frac{15}{4}\mathbf{a} + \frac{5}{4}\mathbf{b}$$

**Original question 29**

4MA1-2H Q17 M01

MODULAR UNIT 2

3 marks

17 Here are two vectors.

$$\vec{FG} = \begin{pmatrix} -5 \\ 2 \end{pmatrix} \quad \vec{HG} = \begin{pmatrix} 4 \\ 14 \end{pmatrix}$$

Calculate the magnitude of the vector  $\vec{HF}$

.....

**Worked solution 29**

Find the magnitude of HF

MODULAR UNIT 2

3 marks

**M01 – Find the magnitude of HF****Find the vector**

$$\overrightarrow{HF} = \overrightarrow{HG} - \overrightarrow{FG} = (4, 14) - (-5, 2) = (9, 12).$$

**Use Pythagoras**

$$|HF| = \sqrt{9^2 + 12^2} = 15.$$

**State the magnitude**

$$|HF| = 15.$$

**Final answer**

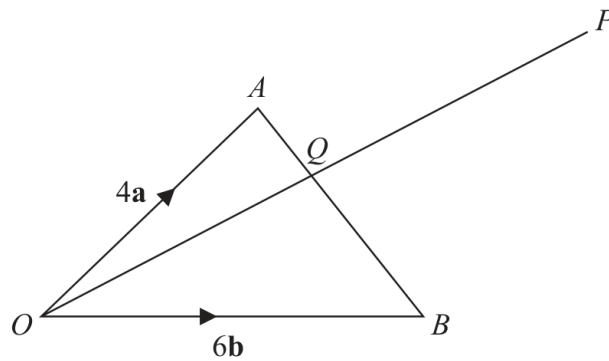
15

**Original question 30**

4MA1-2H Q21 Q21

**ORIGINAL QUESTION****5 marks**

21

Diagram **NOT**  
accurately drawn $OAB$  is a triangle. $Q$  is the point on  $AB$  such that  $OQP$  is a straight line.

$$\vec{OA} = 4\mathbf{a} \quad \vec{OB} = 6\mathbf{b} \quad \vec{AP} = 2\mathbf{a} + 8\mathbf{b}$$

Using a vector method, find the ratio  $AQ:QB$ 

$$AQ:QB = \dots\dots\dots$$

**Worked solution 30**

Chapter: Vectors

MODULAR UNIT 2

5 marks

**Q21 – Chapter: Vectors**

Family: Use vectors with ratios and intersections • Variant: Recover a division ratio using vector parameters

**Method**

$$\mathbf{OP} = \mathbf{OA} + \mathbf{AP} = 6\mathbf{a} + 8\mathbf{b}.$$

Why: Add the consecutive displacement vectors.

**Method**

$$\text{Let } \mathbf{AQ} = t \mathbf{AB}. \text{ Then } \mathbf{OQ} = \mathbf{OA} + t(\mathbf{OB} - \mathbf{OA}) = (4 - 4t)\mathbf{a} + 6t\mathbf{b}.$$

Why: Parameterise Q along AB.

**Method**

$$\text{Since O, Q, P are collinear, } \mathbf{OQ} = \lambda(6\mathbf{a} + 8\mathbf{b}).$$

Why: Parallel vectors are scalar multiples.

**Method**

$$\text{Equating coefficients gives } 4 - 4t = 6\lambda \text{ and } 6t = 8\lambda, \text{ so } t = 8/17.$$

Why: Use independence of a and b to solve the two scalar equations.

**Method**

$$\text{Therefore } \mathbf{AQ}:\mathbf{QB} = (8/17):(9/17) = 8:9.$$

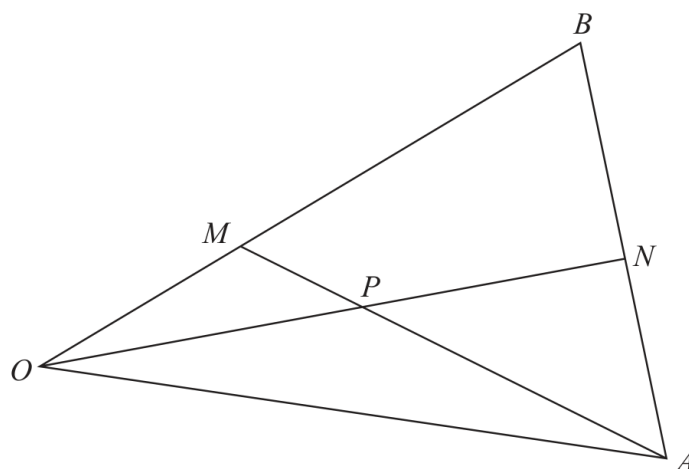
Why: The remaining fraction of AB is  $1 - t$ .

**Final answer**

$$\text{ratio: } \mathbf{AQ} : \mathbf{QB} = 8 : 9$$

**Original question 13**

4MA1-1H Q22

**ORIGINAL QUESTION****5 marks****22** The diagram shows triangle  $OAB$ Diagram **NOT**  
accurately drawn

$$\vec{OA} = 8\mathbf{a} \quad \vec{OB} = 6\mathbf{b}$$

 $M$  is the point on  $OB$  such that  $OM:MB = 1:2$  $N$  is the midpoint of  $AB$  $P$  is the point of intersection of  $ON$  and  $AM$ 

Using a vector method, find  $\vec{OP}$  as a simplified expression in terms of  $\mathbf{a}$  and  $\mathbf{b}$   
 Show your working clearly.

$$\vec{OP} = \dots\dots\dots$$

**Worked solution 13**

Chapter: Vectors

MODULAR UNIT 2

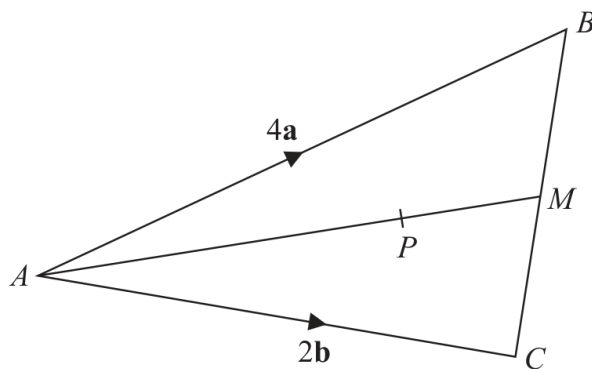
5 marks

**Q22 – Chapter: Vectors**

Family: Use vectors with ratios and intersections • Variant: Find an intersection vector from two paths

**Method** $OM = (1/3)OB = 2b$ , so M has position vector  $2b$ .*Why: Use  $OM:MB = 1:2$ .***Method**N is the midpoint of AB, so  $ON = 4a + 3b$ .*Why: Average  $OA = 8a$  and  $OB = 6b$ .***Method**Let  $OP = \lambda(4a + 3b)$  and also  $OP = 8a + t(2b - 8a)$ .*Why: P lies on both ON and AM.***Method**Comparing coefficients gives  $8 - 8t = 4\lambda$  and  $2t = 3\lambda$ .*Why: Equate the a and b components.***Method**Solving gives  $\lambda = 1/2$ , so  $OP = 2a + 3/2 b$ .*Why: Substitute  $\lambda$  into  $OP = \lambda ON$ .***Final answer****OP:**  $2a + (3/2)b$

23

Diagram **NOT**  
accurately drawn $ABC$  is a triangle.The midpoint of  $BC$  is  $M$ . $P$  is a point on  $AM$ .

$$\vec{AB} = 4\mathbf{a}$$

$$\vec{AC} = 2\mathbf{b}$$

$$\vec{AP} = \frac{3}{2}\mathbf{a} + \frac{3}{4}\mathbf{b}$$

Find the ratio  $AP:PM$

## Worked solution

January 2020 | Question 23 | 3 marks

Family: Use vectors with ratios and intersections • Variant: Recover a division ratio using vector parameters

Method / working

1.  $AB=4a$  and  $AC=2b$ , so  $AM=2a+b$ .
  2.  $AP=(\frac{3}{2})a+(\frac{3}{4})b=(\frac{3}{4})(2a+b)=(\frac{3}{4})AM$ .
  3. Therefore  $PM=(\frac{1}{4})AM$  and  $AP:PM=3:1$ .
- Final answer:  $AP:PM=3:1$

23  $OAB$  is a triangle.

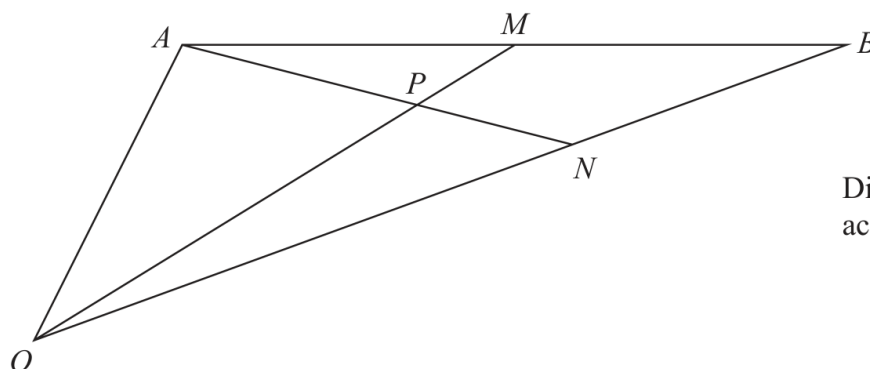


Diagram **NOT**  
accurately drawn

$$\vec{OA} = 2\mathbf{a} \text{ and } \vec{OB} = 2\mathbf{b}$$

$M$  is the midpoint of  $AB$ .

$N$  is the point on  $OB$  such that  $ON:NB = 2:1$

$P$  is the point on  $AN$  such that  $OPM$  is a straight line.

Use a vector method to find  $OP:PM$

Show your working clearly.

## Worked solution

January 2021 | Question 23 | 6 marks

Family: Use vectors with ratios and intersections • Variant: Recover a division ratio using vector parameters

Method / working

1. Write vectors for M and N from  $OA=2a$  and  $OB=2b$ .
2. Express OP along AN and also along OM because O,P,M are collinear.
3. Equate the a and b coefficients.
4. Solve the resulting scalar parameter.
5. Obtain  $OP=(4/5)OM$ .
6. Hence  $PM=(1/5)OM$  and  $OP:PM=4:1$ .

Final answer:  $OP:PM=4:1$

26  $OACB$  is a trapezium.

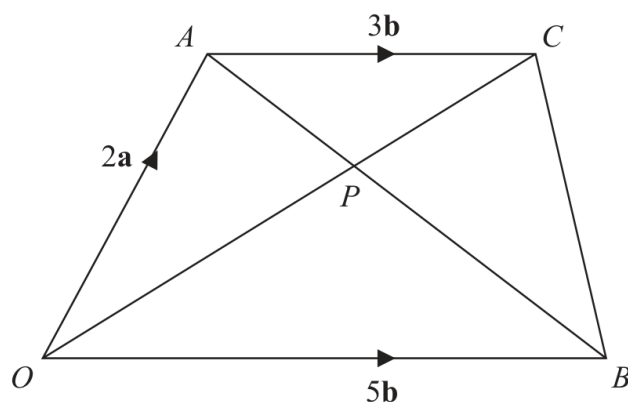


Diagram **NOT**  
accurately drawn

$$\vec{OA} = 2\mathbf{a} \quad \vec{OB} = 5\mathbf{b} \quad \vec{AC} = 3\mathbf{b}$$

The diagonals,  $OC$  and  $AB$ , of the trapezium intersect at the point  $P$ .

Find and simplify an expression, in terms of  $\mathbf{a}$  and  $\mathbf{b}$ , for  $\vec{OP}$   
Show your working clearly.

$$\vec{OP} = \dots\dots\dots$$

## Worked solution

June 2021 | Question 26 | 5 marks

Family: Use vectors with ratios and intersections • Variant: Find an intersection vector from two paths

Method / working

1. Because P lies on OC, write  $OP = n(2a + 3b)$ .
  2. Because P lies on AB, write a second vector equation using  $OA = 2a$  and  $OB = 5b$ .
  3. Equate coefficients of  $a$  and  $b$  to solve the scalar  $n = 5/8$ .
  4. Substitute  $n$  into  $OP = (5/8)(2a + 3b)$ .
  5. Simplify:  $OP = (5/4)a + (15/8)b$ .
- Final answer:  $(5/4)a + (15/8)b$

14  $ABCDEF$  and  $GHIJKL$  are regular hexagons each with centre  $O$ .

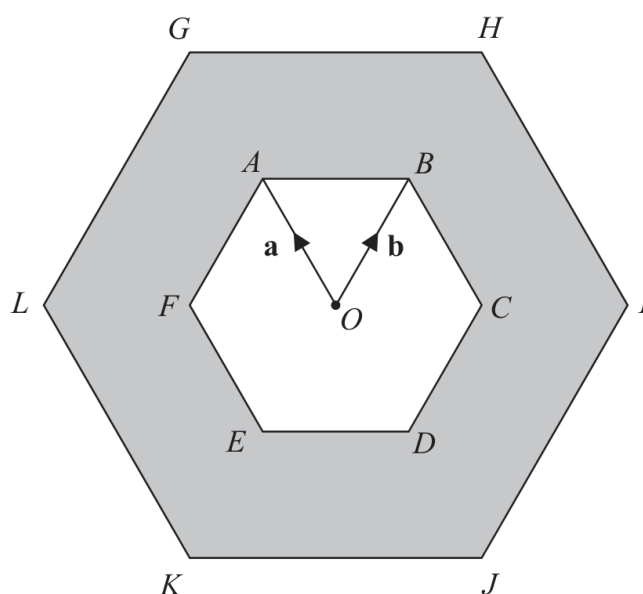


Diagram **NOT**  
accurately drawn

$GHIJKL$  is an enlargement of  $ABCDEF$ , with centre  $O$  and scale factor 2

$$\vec{OA} = \mathbf{a} \quad \vec{OB} = \mathbf{b}$$

(a) Write the following vectors, in terms of  $\mathbf{a}$  and  $\mathbf{b}$ .  
Simplify your answers.

(i)  $\vec{AB}$

(1)

(ii)  $\vec{KI}$

(2)

(iii)  $\vec{LD}$

(2)

The triangle  $OAB$  has an area of  $5 \text{ cm}^2$

(b) Calculate the area of the shaded region.

(3)

$\text{cm}^2$

## Worked solution

November 2021 | Question 14 | 1 marks

Family: Read and represent vectors • Variant: Express a directed segment in basis vectors

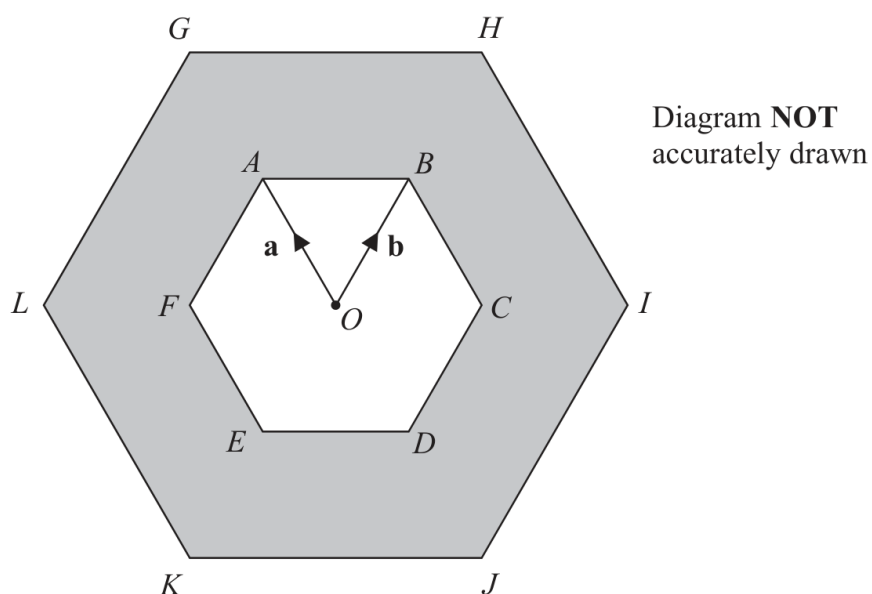
Method / working

1.  $\overrightarrow{OA} = \mathbf{a}$ ,  $\overrightarrow{OB} = \mathbf{b}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

Final answer: (i)  $\mathbf{b} - \mathbf{a}$

14  $ABCDEF$  and  $GHIJKL$  are regular hexagons each with centre  $O$ .



$GHIJKL$  is an enlargement of  $ABCDEF$ , with centre  $O$  and scale factor 2

$$\vec{OA} = \mathbf{a} \quad \vec{OB} = \mathbf{b}$$

(a) Write the following vectors, in terms of  $\mathbf{a}$  and  $\mathbf{b}$ .  
Simplify your answers.

(i)  $\vec{AB}$

(1)

(ii)  $\vec{KI}$

(2)

(iii)  $\vec{LD}$

(2)

The triangle  $OAB$  has an area of  $5 \text{ cm}^2$

(b) Calculate the area of the shaded region.

(3)

$\text{cm}^2$

# Worked solution

November 2021 | Question 14 | 2 marks

Family: Calculate with vectors | Variant: Find a resultant along several moves

## METHOD / WORKING

1. (a)(i)
2.  $\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$
3.  $= \mathbf{b} - \mathbf{a}$
4. For a regular hexagon,
5.  $\overrightarrow{OC} = \mathbf{b} - \mathbf{a}, \quad$
6.  $\overrightarrow{OD} = -\mathbf{a}, \quad$
7.  $\overrightarrow{OE} = -\mathbf{b}, \quad$
8.  $\overrightarrow{OF} = \mathbf{a} - \mathbf{b}$
9. The outer hexagon has scale factor 2, so
10.  $\overrightarrow{OI} = 2(\mathbf{b} - \mathbf{a}), \quad$
11.  $\overrightarrow{OK} = -2\mathbf{b}, \quad$
12.  $\overrightarrow{OL} = 2(\mathbf{a} - \mathbf{b})$

## FINAL ANSWER

(ii)  $4\mathbf{b} - 2\mathbf{a}$

14  $ABCDEF$  and  $GHIJKL$  are regular hexagons each with centre  $O$ .

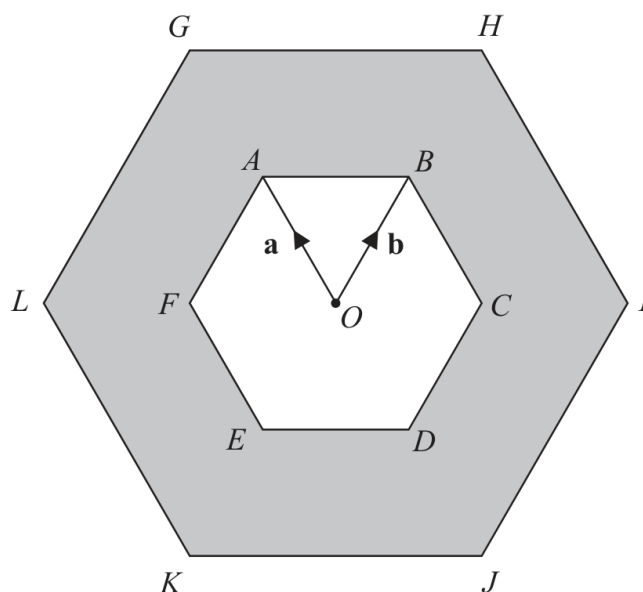


Diagram **NOT**  
accurately drawn

$GHIJKL$  is an enlargement of  $ABCDEF$ , with centre  $O$  and scale factor 2

$$\vec{OA} = \mathbf{a} \quad \vec{OB} = \mathbf{b}$$

(a) Write the following vectors, in terms of  $\mathbf{a}$  and  $\mathbf{b}$ .  
Simplify your answers.

(i)  $\vec{AB}$

(1)

(ii)  $\vec{KI}$

(2)

(iii)  $\vec{LD}$

(2)

The triangle  $OAB$  has an area of  $5 \text{ cm}^2$

(b) Calculate the area of the shaded region.

(3)

$\text{cm}^2$

## Worked solution

November 2021 | Question 14 | 2 marks

Family: Use vectors with ratios and intersections • Variant: Use vectors in a parallelogram, trapezium or polygon path

Method / working

1. (ii)

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

2.  $\overrightarrow{KI}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

3.  $=\overrightarrow{OI}-\overrightarrow{OK}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

4.  $=2(\mathbf{b}-\mathbf{a})-(-2\mathbf{b})$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

5.  $=4\mathbf{b}-2\mathbf{a}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

Final answer: (iii)  $2\mathbf{b}-3\mathbf{a}$

19  $OAB$  is a triangle.

$$\vec{OA} = \mathbf{a} \quad \vec{OB} = \mathbf{b}$$

The point  $C$  lies on  $OA$  such that  $OC : CA = 1 : 2$

The point  $D$  lies on  $OB$  such that  $OD : DB = 1 : 2$

Using a vector method, prove that  $ABDC$  is a trapezium.

(Total for Question 19 is 3 marks)

## Answer reference | May-Nov 2020 | Question 19

Family: Use vectors with ratios and intersections • Variant: Use vectors in a parallelogram, trapezium or polygon path

|    |                                                                                                                                     |                                                                               |   |    |                                                                                                                                                                                                                                     |
|----|-------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|---|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 19 | $\overrightarrow{AB} = -\mathbf{a} + \mathbf{b}$ or $\overrightarrow{BA} = \mathbf{a} - \mathbf{b}$                                 |                                                                               |   | M1 | Correct diagram (condone missing vector labels or arrows – with $C$ on line segment $OA$ and $D$ on line segment $OB$ ) <b>OR</b> for finding $\overrightarrow{AB}$ or $\overrightarrow{BA}$ - may be seen as part of later working |
|    | $\overrightarrow{CD} = \frac{1}{3}(-\mathbf{a} + \mathbf{b})$ or<br>$\overrightarrow{DC} = \frac{1}{3}(\mathbf{a} - \mathbf{b})$ oe |                                                                               |   | M1 | Method to find $\overrightarrow{CD}$ or $\overrightarrow{DC}$                                                                                                                                                                       |
|    |                                                                                                                                     | Correct vectors and conclusion including <u>parallel</u> and <u>trapezium</u> | 3 | A1 | eg $\overrightarrow{AB}$ ( $AB$ ) and $\overrightarrow{CD}$ ( $CD$ ) are parallel therefore $ABDC$ is a trapezium                                                                                                                   |
|    |                                                                                                                                     |                                                                               |   |    | <b>Total 3 marks</b>                                                                                                                                                                                                                |

## Worked solution

May-Nov 2020 | Question 19 | 3 marks

Family: Use vectors with ratios and intersections • Variant: Use vectors in a parallelogram, trapezium or polygon path

Official final mark-scheme pages are embedded immediately after this question.

24  $OAED$  is a quadrilateral.

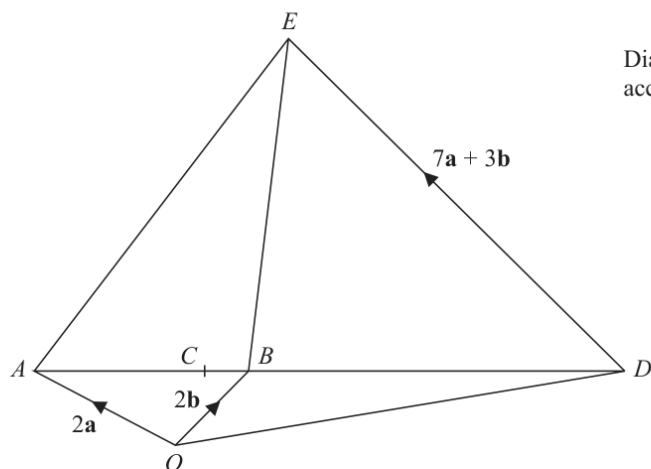


Diagram **NOT**  
accurately drawn

$$\vec{OA} = 2\mathbf{a} \quad \vec{OB} = 2\mathbf{b} \quad \vec{DE} = 7\mathbf{a} + 3\mathbf{b}$$

$$AB : BD = 1 : 2$$

The point  $C$  on  $AB$  is such that  $OCE$  is a straight line.

Use a vector method to find the ratio of  $OC : CE$



(Total for Question 24 is 5 marks)

## Answer reference | January 2023 | Question 24

Family: Use vectors with ratios and intersections • Variant: Find a section or midpoint vector

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |       |   |                                                                                                                                                                                                                                  |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 24 | e.g.<br>$(\overrightarrow{AB}) = 2\mathbf{b} - 2\mathbf{a}$ oe or<br>$(\overrightarrow{BA}) = 2\mathbf{a} - 2\mathbf{b}$ oe or<br>$(\overrightarrow{BD}) = 2(2\mathbf{b} - 2\mathbf{a}) (= 4\mathbf{b} - 4\mathbf{a})$ oe or<br>$(\overrightarrow{AD}) = 3(2\mathbf{b} - 2\mathbf{a}) (= 6\mathbf{b} - 6\mathbf{a})$ oe                                                                                                                                                                                                                               |       | 5 | M1                                                                                                                                                                                                                               |
|    | e.g.<br>$(\overrightarrow{OE}) = 2\mathbf{b} + 2(2\mathbf{b} - 2\mathbf{a}) + 7\mathbf{a} + 3\mathbf{b} (= 3\mathbf{a} + 9\mathbf{b})$ oe or<br>$(\overrightarrow{OC}) = 2\mathbf{a} + \lambda(2\mathbf{b} - 2\mathbf{a}) = (2 - 2\lambda)\mathbf{a} + 2\lambda\mathbf{b}$ oe or $2\mathbf{b} + \lambda(2\mathbf{a} - 2\mathbf{b})$ or<br>$(\overrightarrow{CE}) = (2\mathbf{b} - 2\mathbf{a}) - \lambda(2\mathbf{b} - 2\mathbf{a}) + 2(2\mathbf{b} - 2\mathbf{a}) + 7\mathbf{a} + 3\mathbf{b} = (1 + 2\lambda)\mathbf{a} + (9 - 2\lambda)\mathbf{b}$ |       |   | M2 for 2 correct paths seen<br>M1 for 1 correct path seen<br><br>Any correct path for $OC$ passing through $A$ or $B$ involving a variable                                                                                       |
|    | e.g.<br>$\frac{2 - 2\lambda}{2\lambda} = \frac{1 + 2\lambda}{9 - 2\lambda}$ oe or $\frac{2 - 2\lambda}{2\lambda} = \frac{3}{9}$ oe or $\frac{(1 + 2\lambda)}{(9 - 2\lambda)} = \frac{1}{3}$ oe or $\lambda = \frac{3}{4}$<br>or<br>$(2 - 2\lambda)\mathbf{a} + 2\lambda\mathbf{b} = \mu(3\mathbf{a} + 9\mathbf{b})$ or $\lambda = \frac{3}{4}$ or $\mu = \frac{1}{6}$<br>or<br>$2\mathbf{b} + \lambda(2\mathbf{a} - 2\mathbf{b}) = \mu(3\mathbf{a} + 9\mathbf{b})$ or $\lambda = \frac{1}{4}$ or $\mu = \frac{5}{6}$                                  |       |   | M1 for comparing coefficients of $\mathbf{a}$ and $\mathbf{b}$ for<br><br>$(OC \text{ and } CE)$ or<br>$(OC \text{ and } OE)$ or<br>$(CE \text{ and } OE)$<br><br>$OC$ is a multiple of $OE$<br><br>Two different paths for $OC$ |
|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 1 : 5 |   | A1 dep M2 oe e.g 2 : 10                                                                                                                                                                                                          |
|    | Working required                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |       |   | <b>Total 5 marks</b>                                                                                                                                                                                                             |

## Answer reference | January 2023 | Question 24

Family: Use vectors with ratios and intersections • Variant: Find a section or midpoint vector

|           |                                                                                                                                                                                                                                                                                                                         |       |   |                            |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|---|----------------------------|
| 24<br>ALT | e.g.<br>$(\overrightarrow{AB}) = 2\mathbf{b} - 2\mathbf{a}$ oe or<br>$(\overrightarrow{BA}) = 2\mathbf{a} - 2\mathbf{b}$ oe or<br>$(\overrightarrow{BD}) = 2(2\mathbf{b} - 2\mathbf{a}) (= 4\mathbf{b} - 4\mathbf{a})$ oe or<br>$(\overrightarrow{AD}) = 3(2\mathbf{b} - 2\mathbf{a}) (= 6\mathbf{b} - 6\mathbf{a})$ oe |       | 5 | M1                         |
|           | e.g.<br>$(\overrightarrow{OE}) = 2\mathbf{b} + 2(2\mathbf{b} - 2\mathbf{a}) + 7\mathbf{a} + 3\mathbf{b} (= 3\mathbf{a} + 9\mathbf{b})$ oe                                                                                                                                                                               |       |   | M1                         |
|           | e.g.<br>$(\overrightarrow{AE}) = 2\mathbf{b} - 2\mathbf{a} + 2(2\mathbf{b} - 2\mathbf{a}) + 7\mathbf{a} + 3\mathbf{b} (= \mathbf{a} + 9\mathbf{b})$ oe                                                                                                                                                                  |       |   | M1                         |
|           | $[\overrightarrow{AE} = \lambda \overrightarrow{AD} + \mu \overrightarrow{OE}]$<br>$\mathbf{a} + 9\mathbf{b} = \lambda(6\mathbf{b} - 6\mathbf{a}) + \mu(3\mathbf{a} + 9\mathbf{b})$ oe or<br>$1 = -6\lambda + 3\mu$ oe and $9 = 6\lambda + 9\mu$ oe or or $\lambda = \frac{1}{4}$ or $\mu = \frac{5}{6}$                |       |   | M1                         |
|           |                                                                                                                                                                                                                                                                                                                         | 1 : 5 |   | A1 dep on M2 oe e.g 2 : 10 |
|           | Working required                                                                                                                                                                                                                                                                                                        |       |   | <b>Total 5 marks</b>       |

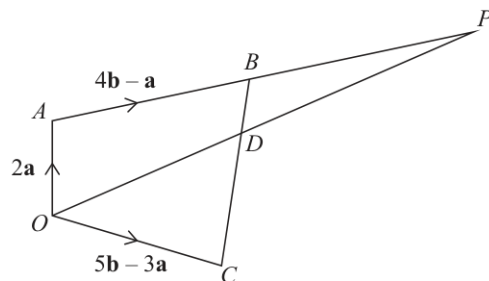
## Worked solution

January 2023 | Question 24 | 5 marks

Family: Use vectors with ratios and intersections • Variant: Find a section or midpoint vector

Official final mark-scheme pages are embedded immediately after this question.

27

Diagram **NOT**  
accurately drawn $OABC$  is a quadrilateral. $ABP$  and  $ODP$  are straight lines.

$$\vec{OA} = 2\mathbf{a} \quad \vec{AB} = 4\mathbf{b} - \mathbf{a} \quad \vec{OC} = 5\mathbf{b} - 3\mathbf{a}$$

- (a) Find an expression in terms of  $\mathbf{a}$  and  $\mathbf{b}$  for the vector  $\vec{BC}$   
Simplify your answer.

(2)

The point  $D$  lies on  $BC$  such that  $BD:DC = 1:3$ Given that  $\vec{OP} = n\vec{OD}$ 

- (b) use a vector method to find the value of  $n$

24



$$n = \dots\dots\dots (4)$$

(Total for Question 27 is 6 marks)

## Worked solution

November 2025 | Question 27 | 6 marks

Family: Use vectors with ratios and intersections • Variant: Find an intersection vector from two paths

Method

$$\vec{OA} = 2\mathbf{a}, \quad \vec{AB} = 4\mathbf{b} - \mathbf{a}, \quad \vec{OC} = 5\mathbf{b} - 3\mathbf{a}$$

First find  $\vec{OB}$ :

$$\vec{OB} = \vec{OA} + \vec{AB}$$

$$\vec{OB} = 2\mathbf{a} + 4\mathbf{b} - \mathbf{a}$$

$$\vec{OB} = \mathbf{a} + 4\mathbf{b}$$

For part (a),

$$\vec{BC} = \vec{OC} - \vec{OB}$$

$$\vec{BC} = (5\mathbf{b} - 3\mathbf{a}) - (\mathbf{a} + 4\mathbf{b})$$

$$\vec{BC} = \mathbf{b} - 4\mathbf{a}$$

For part (b),  $BD:DC = 1:3$ , so

$$\vec{OD} = \vec{OB} + \frac{1}{4}\vec{BC}$$

$$\vec{OD} = \mathbf{a} + 4\mathbf{b} + \frac{1}{4}(\mathbf{b} - 4\mathbf{a})$$

$$\vec{OD} = \frac{3}{4}\mathbf{a} + \frac{17}{4}\mathbf{b}$$

Since A, B, P are collinear, let

$$\vec{OP} = \vec{OA} + s\vec{AB}$$

$$\vec{OP} = 2\mathbf{a} + s(4\mathbf{b} - \mathbf{a})$$

$$\vec{OP} = (2-s)\mathbf{a} + 4s\mathbf{b}$$

But O, D, P are collinear, and  $\vec{OD}$  has only a  $\mathbf{b}$  component.

So

$$2-s=0$$

$$s=2$$

Then

$$\vec{OP} = 8\mathbf{b}$$

Given

$$\vec{OP} = n\vec{OD}$$

$$8\mathbf{b} = n\left(\frac{3}{4}\mathbf{a} + \frac{17}{4}\mathbf{b}\right)$$

$$n = \frac{32}{17}$$

$$\text{Answer: } \vec{BC} = \mathbf{b} - 4\mathbf{a}, \quad n = \frac{32}{17}$$